
UCL COMP0078: Supervised Learning

Solution Sheet

Rademacher Complexity

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Overview of Problems

- **Problem 1:** Analyze the symmetrization argument used to bound the generalization error of a hypothesis in terms of the Rademacher complexity.
- **Problem 2:** Derive the Rademacher complexity for linear predictors.
- **Problem 3:** Demonstrate the connection between Rademacher complexity and covering numbers through a straightforward discretization approach.

Problem 1

In the lecture, it was shown that we could bound the following supremum as follows

$$\sup_{f \in \mathcal{F}} L_D(f) - L_S(f) \leq E_{S'} \left[\sup_{f \in \mathcal{F}} L_{S'}(f) - L_S(f) \right] \quad (1)$$

$$\leq \frac{1}{m} E_{S,S'} \left[\sup_{f \in \mathcal{F}} \sum_{i=1}^m (f(z'_i) - f(z_i)) \right] \quad (2)$$

$$= \frac{1}{m} E_{S,S'} \left[\sup_{f \in \mathcal{F}} \sum_{i=1}^m (f(z'_i) - f(z_i)) \right] \quad (3)$$

$$= \frac{1}{m} E_{S,S',\sigma} \left[\sup_{f \in \mathcal{F}} \sum_{i=1}^m \sigma_i(f(z'_i) - f(z_i)) \right] \quad (4)$$

In this question, we detail the equality from (3) to (4).

1) Prove the following identity

$$\mathbb{E}_{S,S'} \left[\sup_{f \in \mathcal{F}} \left(f(z'_j) - f(z_j) + \sum_{i \neq j} (f(z'_i) - f(z_i)) \right) \right] = \mathbb{E}_{S,S'} \left[\sup_{f \in \mathcal{F}} \left(f(z_j) - f(z'_j) + \sum_{i \neq j} (f(z'_i) - f(z_i)) \right) \right]$$

2) Using the above identity, prove the following identity

$$\mathbb{E}_{S,S'} \left[\sup_{f \in \mathcal{F}} \left(f(z'_j) - f(z_j) + \sum_{i \neq j} (f(z'_i) - f(z_i)) \right) \right] = \mathbb{E}_{S,S',\sigma_j} \left[\sup_{f \in \mathcal{F}} \left(\sigma_j(f(z_j) - f(z'_j)) + \sum_{i \neq j} (f(z'_i) - f(z_i)) \right) \right]$$

3) Using the above identity, prove why the following is true

$$\mathbb{E}_{S,S'} \left[\sup_{f \in \mathcal{F}} \left(f(z'_j) - f(z_j) + \sum_{i \neq j} (f(z'_i) - f(z_i)) \right) \right] = \mathbb{E}_{S,S',\sigma} \left[\sup_{f \in \mathcal{F}} \left(\sum_{i=1}^m \sigma_i(f(z'_i) - f(z_i)) \right) \right]$$

Solution 1

1) Since for each j , z_j and z'_j are i.i.d variables, we can swap them inside the expectation without any issue. Hence, proving the first identity.

2) Applying the result from 1) we obtain the following

$$\begin{aligned}
& \mathbb{E}_{S, S', \sigma_j} \left[\sup_{f \in \mathcal{F}} \left(\sigma_j(f(z_j) - f(z'_j)) + \sum_{i \neq j} (f(z'_i) - f(z_i)) \right) \right] \\
&= \frac{1}{2} \mathbb{E}_{S, S'} \left[\sup_{f \in \mathcal{F}} \left(f(z'_j) - f(z_j) + \sum_{i \neq j} (f(z'_i) - f(z_i)) \right) \right] + \frac{1}{2} \mathbb{E}_{S, S'} \left[\sup_{f \in \mathcal{F}} \left(f(z_j) - f(z'_j) + \sum_{i \neq j} (f(z'_i) - f(z_i)) \right) \right] \\
&= \frac{1}{2} \mathbb{E}_{S, S'} \left[\sup_{f \in \mathcal{F}} \left(f(z'_j) - f(z_j) + \sum_{i \neq j} (f(z'_i) - f(z_i)) \right) \right] + \frac{1}{2} \mathbb{E}_{S, S'} \left[\sup_{f \in \mathcal{F}} \left(f(z_j) - f(z'_j) + \sum_{i \neq j} (f(z'_i) - f(z_i)) \right) \right] \\
&= \mathbb{E}_{S, S'} \left[\sup_{f \in \mathcal{F}} \left(f(z'_j) - f(z_j) + \sum_{i \neq j} (f(z'_i) - f(z_i)) \right) \right]
\end{aligned}$$

3) Since we choose an arbitrary j , this is true for any datapoint in our sample. Therefore, we can add independent Rademacher variable in front for each of each $f(z'_i) - f(z_i)$. Hence,

$$\mathbb{E}_{S, S'} \left[\sup_{f \in \mathcal{F}} \left(f(z'_j) - f(z_j) + \sum_{i \neq j} (f(z'_i) - f(z_i)) \right) \right] = \mathbb{E}_{S, S', \sigma} \left[\sup_{f \in \mathcal{F}} \left(\sum_{i=1}^m \sigma_i (f(z'_i) - f(z_i)) \right) \right]$$

Problem 2

Suppose k is a bounded kernel such that $\sup_x \sqrt{k(x, x)} = \kappa < \infty$ and there exists an explicit feature map $\phi : \mathcal{X} \rightarrow \mathbb{R}^n$ where $k(x, y) = \phi^T(x)\phi(y)$.

1) Give an example of a kernel k such that $\forall x, y \in \mathcal{X}, \sqrt{k(x, y)} \leq \kappa$

2) Let $\gamma > 0$ be fixed constant value, then for any sample $S = (X_1, \dots, X_n)$ we can bound the empirical Rademacher complexity as follows

$$\hat{\mathcal{R}}_S(B_k(\gamma)) \leq \frac{\gamma \sqrt{\text{trace}(K)}}{n}, \quad \mathcal{R}_n(B_k(M)) \leq \frac{\gamma \kappa}{\sqrt{n}}$$

where K is the kernel matrix for the sample S and $B_k(\gamma) = \{f_w(\mathbf{x}) \mid \|w\|_2 \leq \gamma, f(\mathbf{x}) = w^T \phi(\mathbf{x})\}$.

3) Using McDiarmid inequality, prove there exists a constant $C_1 > 0$

$$\mathcal{R}_n(B_k(\gamma)) \leq \hat{\mathcal{R}}_S(B_k(\gamma)) + C_1 \sqrt{\frac{2 \log(2/\delta)}{n}}$$

holds with at least $1 - \delta$ probability.

4) Prove that there exists constant $C_1, C_2 > 0$ such that the following upper bound for the expected loss for kernel ridge regression

$$\mathbb{E} [\ell(h(x), y)] \leq \frac{1}{n} \sum_{i=1}^n \ell(h(x_i), y_i) + \frac{C_1 \sqrt{\text{trace}(K)}}{n} + C_2 \sqrt{\frac{\log(2/\delta)}{n}}$$

holds with probability at least $1 - \delta$ for a c -bounded μ -Lipschitz loss function and when $(x_i, y_i)_{i=1}^n$ is just a sample on size m from the distribution $P(x, y)$.

Solution 2

1) There are numerous examples, a simple trivial example would be the constant kernel. A more non-trivial and widely used kernel is the gaussian kernel.

2)

$$\begin{aligned} \hat{\mathcal{R}}_S(B_k(\gamma)) &= \frac{1}{m} \mathbb{E}_{\sigma} \left[\sup_{f \in B_k(\gamma)} \sum_{i=1}^n \sigma_i f(z_i) \right] \\ &= \frac{1}{m} \mathbb{E}_{\sigma} \left[\sup_{w \in \{w \in \mathbb{R}^n \mid \|w\|_2 \leq \gamma\}} \sum_{i=1}^n \sigma_i w^T \phi(z_i) \right] \\ &= \frac{1}{m} \mathbb{E}_{\sigma} \left[\sup_{w \in \{w \in \mathbb{R}^n \mid \|w\|_2 \leq \gamma\}} \left\langle w, \sum_{i=1}^n \sigma_i \phi(z_i) \right\rangle \right] \end{aligned}$$

From Cauchy-Swartz inequality we know that for any $w \in \{w \in \mathbb{R}^n \mid \|w\|_2 \leq \gamma\}$

$$\left\langle w, \sum_{i=1}^n \sigma_i \phi(z_i) \right\rangle \leq \|w\| \left\| \sum_{i=1}^n \sigma_i \phi(z_i) \right\| \leq \gamma \left\| \sum_{i=1}^n \sigma_i \phi(z_i) \right\|$$

Thus,

$$\sup_{w \in \{w \in \mathbb{R}^n \mid \|w\|_2 \leq \gamma\}} \left\langle w, \sum_{i=1}^n \sigma_i \phi(z_i) \right\rangle \leq \gamma \left\| \sum_{i=1}^n \sigma_i \phi(z_i) \right\| \quad (5)$$

Therefore, from (5) we conclude that

$$\begin{aligned} \hat{R}_S(B_k(\gamma)) &= \frac{1}{m} \mathbb{E}_\sigma \left[\sup_{w \in \{w \in \mathbb{R}^n \mid \|w\|_2 \leq \gamma\}} \left\langle w, \sum_{i=1}^n \sigma_i \phi(z_i) \right\rangle \right] \\ &\leq \frac{\gamma}{n} \mathbb{E}_\sigma \left[\sqrt{\left\| \sum_{i=1}^n \sigma_i \phi(z_i) \right\|^2} \right] \\ &\leq \frac{\gamma}{n} \sqrt{\mathbb{E}_\sigma \left[\left\| \sum_{i=1}^n \sigma_i \phi(z_i) \right\|^2 \right]} \text{ by Jensen inequality} \\ &= \frac{\gamma}{n} \sqrt{\mathbb{E}_\sigma \left[\sum_{i=1}^n \phi(z_i)^T \phi(z_i) + \sum_{i \neq j} \sigma_i \sigma_j \phi(z_i)^T \phi(z_j) \right]} \\ &= \frac{\gamma}{n} \sqrt{\mathbb{E}_\sigma \left[\sum_{i=1}^n \phi(z_i)^T \phi(z_i) \right]}, \text{ as } \mathbb{E}_\sigma[\sigma_i \sigma_j] = 0 \\ &= \frac{\gamma}{n} \sqrt{\mathbb{E}_\sigma \left[\sum_{i=1}^n k(z_i, z_i) \right]} \\ &= \frac{\gamma \sqrt{\text{trace}[K]}}{n} \leq \frac{\gamma \kappa}{\sqrt{n}} \text{ as } k(x, y) \leq \kappa \text{ for any } x, y \in \mathcal{X} \end{aligned}$$

Since γ and κ are constant with respect to the samples taken, $R_n(B_k(\gamma)) \leq \frac{\gamma \kappa}{\sqrt{n}}$.

Extra exercise: Justify why the inequality in (5) can be replaced with an equality.

3) Let $g(z_1, \dots, z_n) = \mathbb{E}_\sigma[\frac{1}{n} \sup_{f \in B_k(\gamma)} \sum_{i=1}^n \sigma_i f(z_i)]$. Then observe that

$$\begin{aligned} |g(z_1, \dots, z_i, \dots, z_n) - g(z_1, \dots, z'_i, \dots, z_n)| &\leq \frac{1}{n} \sup_{f \in B_k(\gamma)} |f(z_i) - f(z'_i)| \\ &= \frac{1}{n} \sup_{w \in \{w \in \mathbb{R}^n \mid \|w\|_2 \leq \gamma\}} |w^T (\phi(z_i) - \phi(z'_i))| \\ &\leq \frac{\gamma}{n} \|\phi(z_i) - \phi(z'_i)\| \leq \frac{\gamma}{n} (\|\phi(z_i)\| + \|\phi(z'_i)\|) \leq \frac{2\gamma\kappa}{n} \end{aligned}$$

Thus, we can use McDiarmid inequality, to conclude that

$$\begin{aligned} \mathbb{E}_S[\mathbb{E}_\sigma[\frac{1}{n} \sup_f \sum_{i=1}^n \sigma_i f(z_i)]] - \mathbb{E}_\sigma[\frac{1}{n} \sup_f \sum_{i=1}^n \sigma_i f(z_i)] &\leq \frac{2\gamma\kappa}{n} \sqrt{\frac{2 \log(2/\delta)}{n}} \\ \implies \mathbb{E}_S[\mathbb{E}_\sigma[\frac{1}{n} \sup_f \sum_{i=1}^n \sigma_i f(z_i)]] &\leq \mathbb{E}_\sigma[\frac{1}{n} \sup_f \sum_{i=1}^n \sigma_i f(z_i)] + \frac{2\gamma\kappa}{n} \sqrt{\frac{2 \log(2/\delta)}{n}} \\ \implies \mathbb{R}_n(B_k(\gamma)) &\leq \hat{\mathbb{R}}_S(B_k(\gamma)) + C_1 \sqrt{\frac{2 \log(2/\delta)}{n}} \end{aligned}$$

where $C_1 = \frac{2\gamma\kappa}{n}$.

4) Substituting the result from 2) to 2) and using the contraction lemma in the lecture, gives us 4)

Problem 3

Let $\mathcal{F} \subset L^2(\mathcal{X})$ be a family of functions such that $\mathcal{F}$ is a compact subset of $L^2(\mathcal{X})$,

1) Write down the definition of a compact set. From the definition of a compact set, what can we say about the existence of $\mathcal{N}(\mathcal{F}, \epsilon, \|\cdot\|)$.

2) In CW2 you have shown Massart lemma (recall part I of CW2), which states that for a finite set of vectors $\mathcal{A} = \{\mathbf{a}_1, \dots, \mathbf{a}_N\}$ in $\mathbb{R}^m$ the Rademacher complexity for this finite set is bounded by

$$\hat{R}_S(\mathcal{A}) \leq \max_{\mathbf{a} \in \mathcal{A}} \|\mathbf{a} - \bar{\mathbf{a}}\| \frac{\sqrt{2 \log N}}{n}$$

where $\bar{\mathbf{a}} = \frac{1}{N} \sum_{i=1}^N \mathbf{a}_i$ and sample set $S = \{z_1, \dots, z_n\}$. Using this fact prove that for a compact set $\mathcal{F}$ there exist $C > 0$ such that

$$\hat{R}_S(\mathcal{F}) \leq \epsilon + C \sqrt{\frac{2 \log \mathcal{N}(\mathcal{F}, \epsilon, \|\cdot\|_{2,S})}{n}}$$

for any $\epsilon > 0$, which is the radius for each ball of our open cover of $\mathcal{F}$, where $\|f\|_{2,S} = \frac{1}{n} \sum_{i=1}^n f(z_i)^2$

Solution 3

1) The subset $\mathcal{F}$ is compact if for every open cover of $\mathcal{F}$ has a finite subcover. Since for every open cover has a finite subcover, $\mathcal{N}(\mathcal{F}, \epsilon, \|\cdot\|) < \infty$.

2) Since $\mathcal{F}$ is compact, there exist an open cover $\bigcup_{i=1}^n B_\epsilon(f'_i, \|\cdot\|_{2,S})$ such that f'_i is the center of each ball and $f'_i \in \mathcal{F}$ (recall all compact set are totally bounded). Given $f \in \mathcal{F}$ there exist $f' \in \mathcal{A} = \{f'_i\}_{i=1}^n$ where $f' \in \arg \min_{\mathcal{A}} \|f'_i - f\|_{2,S}$, observe that

$$\sum_{i=1}^n \sigma_i(f(z_i) - f'(z_i)) \leq \sum_{i=1}^n |f(z_i) - f'(z_i)| \leq n \sqrt{\frac{1}{n} \sum_{i=1}^n (f(z_i) - f'(z_i))^2} \leq n\epsilon$$

Then,

$$\begin{aligned} \hat{R}_S(\mathcal{F}) &= \mathbb{E}_\sigma \left[\sup_f \frac{1}{n} \sum_{i=1}^n \sigma_i f(z_i) \right] \\ &= \mathbb{E}_\sigma \left[\sup_f \frac{1}{n} \sum_{i=1}^n \sigma_i (f'(z_i) + (f - f')(z_i)) \right] \\ &\leq \mathbb{E}_\sigma \left[\sup_f \frac{1}{n} \sum_{i=1}^n \sigma_i f'(z_i) + \sup_f \frac{1}{n} \sum_{i=1}^n \sigma_i (f - f')(z_i) \right] \\ &\leq \mathbb{E}_\sigma \left[\sup_f \frac{1}{n} \sum_{i=1}^n \sigma_i f'(z_i) + \epsilon \right] \\ &\leq \epsilon + \mathbb{E}_\sigma \left[\sup_{f \in \mathcal{A}} \frac{1}{n} \sum_{i=1}^n \sigma_i f(z_i) \right] \leq \epsilon + C \frac{\sqrt{2 \log \mathcal{N}(\mathcal{F}, \epsilon, \|\cdot\|_{2,S})}}{n} \end{aligned}$$

where $C = \max_{\mathbf{a} \in \mathcal{A}} \|\mathbf{a} - \bar{\mathbf{a}}\|$